

MICA: Can an AI Understand What a Minecraft Builder Is Doing?

Minecraft Intention-Aware Collaborative Agent · Benjamin [surname], [institution] · recognizing situations and intentions from raw observation — no instructions, no privileged game state

Why this matters

In structured work — construction, warehouses, assembly — an assistant that must be **told everything is bottlenecked by the telling**. The scarce resource is the human’s attention. An assistant that *infers* what you are doing removes that bottleneck.

Assistance-game research formalized this (CIRL [1]) and proved goal inference is the load-bearing part: a **confidently wrong helper is worse than none** (Watch-And-Help [2]: 40% of episodes, the helper undid the human’s work).

The gap: every prior assistant reads oracle game state. **Robots don’t get oracle state**. MICA asks the question under robot conditions — perception in, calibration measured, authority earned.

Technical method: the pipeline

Capture — Fabric mod, 20 Hz proof-grade
every block event actor-attributed · POV frames · input state · hotbar/inventory

Perception — frozen 2D + 3D encoders
VPT trunk (70k h of play) → behavior embedding h_{2d} · MineCLIP → 5-goal similarity s_{goal}
voxel replay → template features · frozen Uni3D-B → shape embedding h_{3d}

Belief — recursive Bayes $b(goal, mode)$
likelihood heads TRAINED on 12,045 real+scripted pairs · resample kernels, ϵ -floor
do-operator: the agent’s own blocks are never evidence

Decoder — 17.5M transformer, 87-token action grammar
TRAINED: next-token warm-start → multi-token retrofit ($n=8$) · real-session pretraining, label-free

Safety gate + FSM — hand-coded, inspectable
observe < suggest < preview < place · frozen thresholds · hysteresis · proximity vetoes

Embodied agent — mineflayer body
follows, scans, voices offers · places ONE block only on your chat “yes”

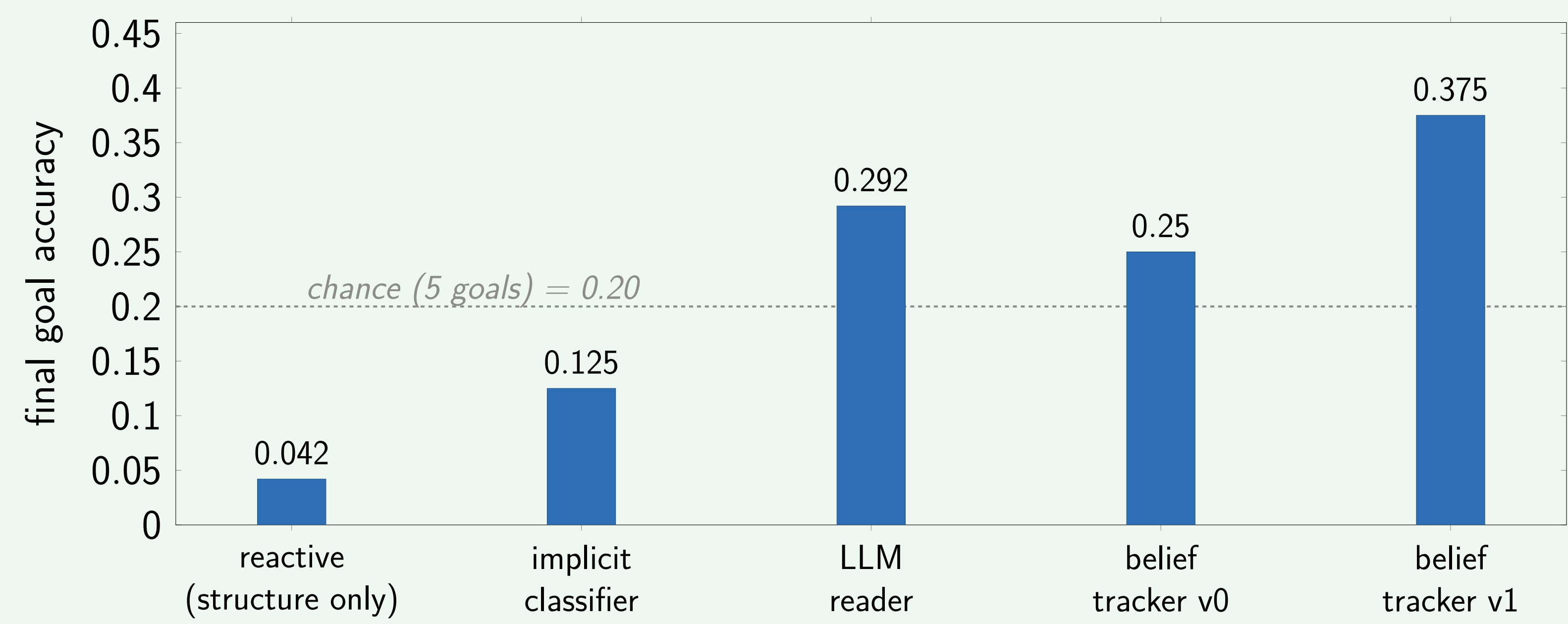
Trained: likelihood heads + decoder (from my sessions). **Frozen:** every foundation encoder [3] — models built to make agents *act* are read backwards, as *evidence about a human*. **Hand-coded:** everything that must be inspectable (gate, consent, rules).

The dataset (all real, all mine)

labeled real free-build sessions	37
matcher-agreed / contested / discarded	10 / 18 / 9
never-seen evaluation sessions	24
held-out real rows (anticipation eval)	1,158
training pairs (one reproducible cascade)	12,045
scripted known-truth corpus	120 builds

One command regenerates every number on this poster from raw recordings.

Finding 1 — Intent inference is NECESSARY off-template



On **24 real sessions no model ever trained on**, the helper that skips intent inference collapses to **0.042** — a **9×** gap to the belief tracker. Exactly where builds stop matching templates, structure lookup stops working. *Intent inference is what survives*.

Finding 2 — It anticipates your next moves, with minimal context

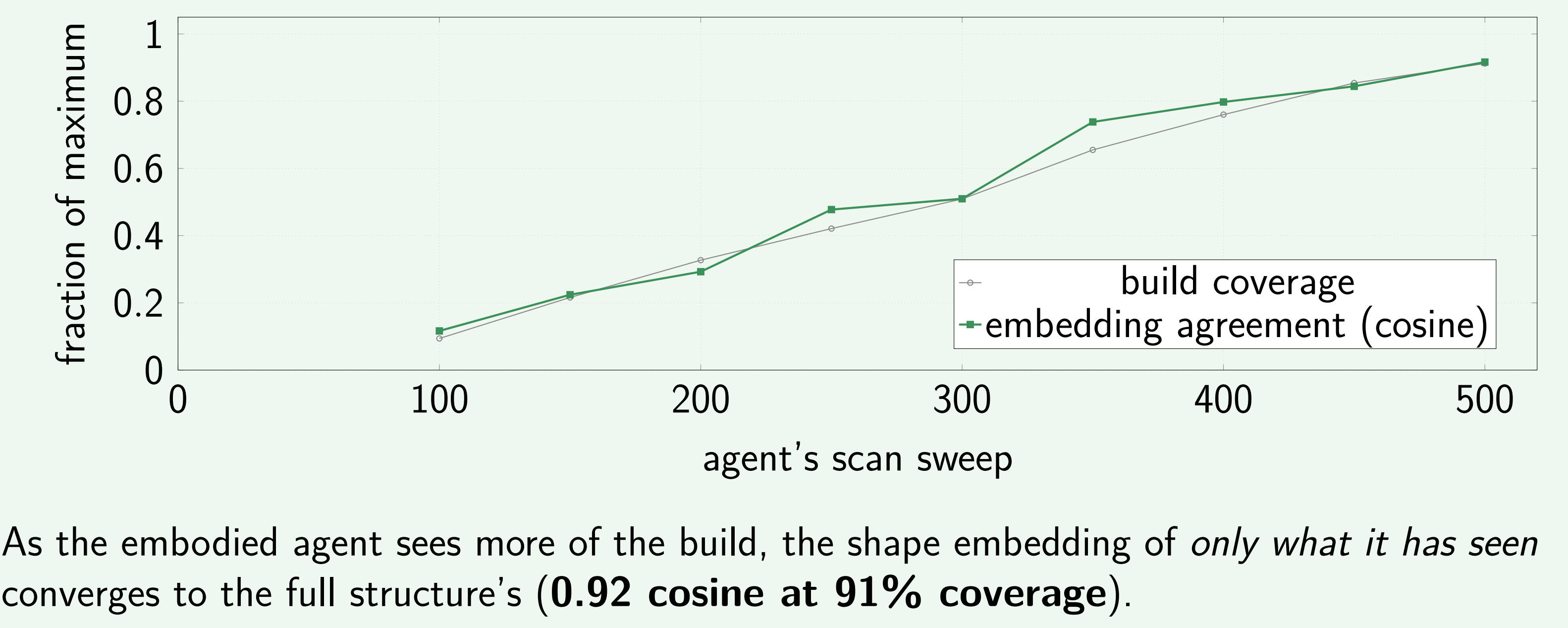
71%
next action
(real sessions)

59%
eight actions ahead
(real sessions)

1.2%
chance level
(87-token grammar)

No instructions, no goal ever specified, frozen perception only. And the training data was free: **label-free pretraining on real sessions** closed a 1.25-nat domain gap (−37% prediction error) — contested and discarded sessions included, because no label enters the sample.

Finding 3 — Understanding accumulates with observation



What failed — and why that is the point

- This project pre-registered its criteria and let them fire:
- ▶ **Early recognition: unproven.** The pre-registered early lock-in test fails for every arm ($p = 0.06$ –1.0). Recognition is mid-build, not early — more sessions are the path.
 - ▶ **Confidence: not calibrated.** Above 0.4 stated confidence, held-out reliability *inverts*. Consequence taken seriously: the confidence-licensed placement route is disabled by its own threshold.
 - ▶ **The belief is not prediction fuel.** Forcing the belief into the action decoder changes nothing ($|\Delta| \leq 0.001$ at every horizon) — its value is *judgment* (what to say, when to act), not forecasting.
 - ▶ **The agent has never placed a block.** Zero, ever — pinned. Assistance here is *designed and safety-gated, never evaluated*. A helper that declines to act without license is the safety design working.
 - ▶ **One verdict was reversed** — by finding a feature leak, repairing the corpus, and re-running the pre-registered experiment (it then won on every criterion). The retraction is on the record.

The system improves lawfully with the right data

Two pre-registered experiments rode one retrain; both won; both adoptions were separate, dated decisions:

	control	adopted
real-session prediction error (NLL)	3.42	2.17
clean-eval accuracy (+ contested labels)	0.00	0.33
calibration error (ECE)	0.258	0.232
earliness (sustained-from, lower=earlier)	0.990	0.897

What this means beyond Minecraft

The approach transfers to any domain with: **(1)** discrete observable actions, **(2)** goals legible from partial structure, **(3)** cheap dense observation of a shared workspace — bricklaying, timber framing, warehouse picking, lab bench work.

Next: recalibration (the gate to autonomy) · consent-route sessions with *placement survival* as the outcome measure · more sessions for the earliness question.

Take one sentence home

“A helper that only pattern-matches the environment dies the moment work deviates from templates — and real work always deviates.
Inferring intention is what survives.”

METHOD — how every number here was made trustworthy

- Pre-registered.** Criteria, decision rules, and holdouts frozen and dated *before* every run — and they fired: one verdict reversed, one table retracted, failures banked.

Known-truth anchored. A scripted corpus whose labels are true by construction: deterministic components are *measured* (matcher 1.0), never assumed.

Leakage rules, standing. Evidence strictly precedes the action it scores; the agent’s own blocks are never evidence (do-operator); violations are asserts, not reviews.

Controlled arms. Four arms differ *only* in the intent mechanism — same gate, same exposure-verified splits; experiments train to separate files, production checksummed.

One-command reproducible. A single ledger-stamped cascade regenerates every number on this poster from the raw recordings.

[1] Hadfield-Menell et al., *Cooperative Inverse Reinforcement Learning*, NeurIPS 2016 · [2] Puig et al., *Watch-And-Help*, ICLR 2021 · [3] Baker et al., *VPT*, NeurIPS 2022 + Fan et al., *MineDojo/MineCLIP*, NeurIPS 2022